

UI/UX Specification Document

Dynamic Train Arrival Prediction Engine (DTAPE)

Document Identifier: UIUX-IR-DTAPE-2026-V1.0
Target Platform: Indian Railways Unified Operational & Public Interface Ecosystem
Design Standard: Minimal Official Modernism (GovTech Clean Line System)
Default Theme: Light Mode (High-Legibility Accessible Baseline)

1. Executive Summary & Design Vision

The Dynamic Train Arrival Prediction Engine (DTAPE) requires a user interface that bridges the operational rigor of Indian Railways control systems with the accessibility of modern consumer digital products.

Traditional railway operations software often suffers from visual clutter, high information density, and disparate visual conventions. DTAPE introduces **Minimal Official Modernism**:

- **Minimalist & Distraction-Free:** Eliminates decorative gradients, skeuomorphic bevels, and heavy drop shadows in favor of crisp borders, purposeful whitespace, and precise typographic contrast.
- **Official & Authoritative:** Incorporates Indian Railways deep heritage blues and clean institutional slates, conveying statutory trust, public safety, and operational reliability.
- **Modern & Dynamic:** Uses subtle micro-interactions, responsive CSS grid architectures, live-updating telemetry pins, and smooth vector rendering to deliver real-time data without cognitive friction.

2. Cognitive Load Reduction & Applied UX Frameworks

Railway controllers, station staff, and passengers make time-sensitive decisions under stress. The user experience directly enforces core cognitive psychology and human-computer interaction (HCI) laws to minimize mental strain.

2.1 Cognitive Load Management Matrix

UX / HCI Principle	Cognitive Risk Addressed	DTAPE UI Implementation Strategy
Hick’s Law	Decision paralysis caused by hundreds of active trains on a controller's screen.	Contextual Layering: Controllers see only high-level train pins and delay statuses by default. Deep telemetry, signal aspects, and conformal prediction bands appear only upon selection or when automatic conflict warnings are triggered.

Miller's Law (7 ± 2 Rule)	Short-term memory overload when analyzing train telemetry.	Information Chunking: Train telemetry is split into four distinct visual cards: (1) Running Status & Speed, (2) Current vs. Booked Schedule, (3) Predicted Delay & Arrival Window, and (4) Recovery Factor Breakdown.
Progressive Disclosure	Overwhelming end passengers with complex mathematical confidence intervals and signaling blocks.	Tiered Discovery: Passengers immediately see the definitive predicted arrival time (e.g., 17:42) and delay badge (+12 min). Tapping "Prediction Details" reveals the confidence interval (17:38 – 17:46) and operational context (e.g., <i>Caution order near Kanpur</i>).
Fitts's Law	Misclicks and slow navigation during critical operational dispatching.	Target Optimization: Critical dispatch buttons, emergency markers, and filter controls have large hit targets ($\geq 44 \text{ pt}$ on mobile, $\geq 36 \text{ px}$ on desktop) placed at prime screen anchor zones (bottom sheets on mobile, persistent top bars on desktop).
Jakob's Law	Learning curves for legacy operators accustomed to mechanical charts and COA layouts.	Mental Model Continuity: Retains standard Indian Railways nomenclature (STA, STD, NSRT, TSR, Loop Line) and preserves the traditional left-to-right station progression and

		linear string-chart conventions of physical control sheets.
Gestalt Principles (Proximity & Region)	Ambiguity over which data point applies to which train or station.	Contained Cards & Enclosures: Related parameters (e.g., scheduled time, actual time, recovery margin) are grouped within distinct single-pixel bordered containers with 8px internal padding.

3. Design System & Visual Foundation

The design system is built entirely on a clean, high-contrast **Light Theme** by default, with strict tokens governing colors, type, spacing, and elevation.

3.1 Color Palette (Default Light Theme)

All colors meet WCAG 2.1 Level AAA contrast ratios (7:1 for normal body text, 4.5:1 for heavy interface elements) against white and neutral backgrounds.

Brand & Neutral Tokens

Token Name	Hex Code	Purpose & Application
--surface-canvas	#F8FAFC	Main application background (clean off-white slate).
--surface-card	#FFFFFF	Background for panels, modal sheets, and data cards.
--surface-subtle	#F1F5F9	Hover states, table headers, and inactive tab pills.
--border-subtle	#E2E8F0	Dividers, structural borders, card perimeters (1px).
--border-strong	#CBD5E1	Focused inputs, active

		container outlines.
--text-primary	#0F172A	Primary titles, train numbers, critical metrics (Deep Slate).
--text-secondary	#475569	Explanatory labels, station names, table column headers.
--text-muted	#94A3B8	Timestamps, static captions, disabled actions.
--brand-ir-navy	#0F3875	Official Indian Railways deep navy (primary identity).
--brand-ir-blue	#1E5AA8	Primary interactive buttons, navigation highlights.
--brand-focus	#3B82F6	Keyboard focus rings, selected state borders.

Operational & Semantic Status Tokens

Semantic State	Fill Token	Border Token	Text Token	Operational Meaning
On Time / Early	#ECFDF5	#A7F3D0	#065F46	Delay ≤ 5 min; running within engineering buffer.
Minor Delay	#FFFBEB	#FDE68A	#92400E	Delay between 6 and 15 min; absorbed by upstream TRT.

Significant Delay	#FFF7ED	#FED7AA	#9A3412	Delay between 16 and 60 min; priority conflicts possible.
Critical Delay	#FEF2F2	#FECACA	#991B1B	Delay \$> 60\$ min; crew relief & platform re-berthing required.
Active Telemetry	#EFF6FF	#BFDDBFE	#1E40AF	Live GAGAN/RTIS satellite signal locked and validated.

3.2 Typography System

The typography uses **Inter** (or **Public Sans**) for standard UI elements and **JetBrains Mono** for numerical telemetry to ensure clear readability:

- **Primary Font Family:** Inter, -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif
- **Monospace Font Family:** "JetBrains Mono", "SF Mono", Menlo, Consolas, monospace

Type Hierarchy

Style Token	Font Size	Line Height	Weight	Font Family	Usage
Display-1	32px	40px	700 (Bold)	Inter	Dashboard summary titles, top-level station arrival boards.
Heading-1	24px	32px	600 (SemiBold)	Inter	Section headers, modal

					panel titles, train name display.
Heading-2	18px	24px	600 (SemiBold)	Inter	Card titles, station names in timeline trackers.
Body-Regular	14px	20px	400 (Regular)	Inter	Standard UI text, descriptions, table rows, form inputs.
Body-Medium	14px	20px	500 (Medium)	Inter	Active buttons, navigation links, prominent labels.
Caption	12px	16px	500 (Medium)	Inter	Status badges, timestamps , kilometric markers.
Mono-Metric	16px	20px	600 (SemiBold)	JetBrains Mono	Live speed (km/h), delay duration, scheduled arrival times.
Mono-Micro	12px	16px	500 (Medium)	JetBrains Mono	Train IDs, platform

					numbers, GPS lat/long values.
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3.3 Iconography System

Iconography follows an outline style (stroke width: 1.5px, optical square: 20x20px or 24x24px) to keep the interface light and legible.

- **Style Standard:** Vector SVG, geometric, rounded terminal caps.
- **Core Icon Mappings:**
 - Train: Train identification, service class indicator.
 - Clock: Timetabled schedule, dwell duration.
 - Activity: Real-time predicted arrival time, ML dynamic inference.
 - AlertTriangle: Temporary Speed Restrictions (TSR), caution orders.
 - CloudFog: Fog-PASS activated, adverse weather visibility restriction.
 - ShieldCheck: Conformal prediction confidence interval verified ($\geq 80\%$).
 - MapPin: Station stop, intermediate reporting point.
 - GitBranch: Loop-line diversion, track turnout, overtake maneuver.
 - Gauge: Instantaneous speed telemetry (km/h).
 - Layers: Platform allocation, yard throat occupancy.

3.4 Spatial Grid, Spacing, & Elevation System

- **Base Unit:** 8-point spatial grid (\$space-1 = 4px\$, \$space-2 = 8px\$, \$space-3 = 12px\$, \$space-4 = 16px\$, \$space-6 = 24px\$, \$space-8 = 32px\$).
- **Card Borders:** 1px solid var(--border-subtle) (#E2E8F0).
- **Corner Radii:**
 - Small Badges / Chips: 4px
 - Buttons / Form Controls: 6px
 - Containers & Cards: 8px
 - Modal Sheets & Dialogs: 12px
- **Elevation & Shadows:**
 - Level 0 (Flat Cards): box-shadow: none; border: 1px solid #E2E8F0;
 - Level 1 (Dropdowns, Popovers): box-shadow: 0 1px 3px 0 rgba(15, 23, 42, 0.08);
 - Level 2 (Modals, Overlays): box-shadow: 0 10px 15px -3px rgba(15, 23, 42, 0.1);

4. Responsive Layout Architecture

The user interface adapts across four primary display targets to support different operational contexts:

Large Control Wall / Dual 4K	→ Multi-Corridor Time-Distance String View
Desktop Workstation (1080p)	→ 3-Pane Controller Console (List + Chart + Inspector)
Tablet / Toughpad (10")	→ 2-Pane Station Master / Platform Turnaround View
Mobile Smartphone (6")	→ 1-Column Passenger / Field Inspector View

4.1 Responsive Breakpoints

Breakpoint Tier	Screen Width Range	Target Environment & Device	Layout Structure
Mobile (xs, sm)	$320\text{ px} - 767\text{ px}$	Passenger mobile browsers, field crew handhelds.	Single-column linear stack, sticky status header, swipeable bottom sheets.
Tablet (md)	$768\text{ px} - 1023\text{ px}$	Station Master tablets, yard inspector devices.	Two-column split (master station overview + platform allocation cards).
Desktop (lg, xl)	$1024\text{ px} - 1919\text{ px}$	Divisional Train Controller desks, HQ consoles.	Three-pane layout (left collapsible train roster, center interactive control chart, right detail drawer).
Ultra-Wide (2xl)	$\geq 1920\text{ px}$	Control room video walls, divisional theater displays.	Multi-corridor simultaneous time-distance charts and real-time conflict feeds.

4.2 Cross-Device Adaptability Rules

A. Mobile Viewport Adaptations

- Collapses complex multi-column data tables into touch-friendly summary cards.

- Replaces dense horizontal time-distance charts with vertical station milestone lists.
- Navigation moves to a fixed bottom navigation bar with thumb-accessible targets (48 px height).

B. Desktop Viewport Adaptations

- Multi-pane split screen allows controllers to view their complete section simultaneously without tab switching.
- High-density tables utilize monospace typography with keyboard shortcuts for search (/), filter (Ctrl+F), and alert acknowledgment (Space).

C. Station Platform Display (PIDS Wall Mount)

- Operates in a high-contrast mode visible from 20 meters away.
- Auto-advances across upcoming trains every 12 seconds with legible display typography (48 px - 64 px).

5. Micro-Interactions, Motion Design, & Animation

Animations are designed to provide clear operational feedback while avoiding visual distractions (150 ms - 250 ms duration).

5.1 Motion Specifications

Interaction Event	Animation Behavior	Duration	Easing Curve	Purpose
Live Telemetry Ping	Green dot scale pulse (1.0 to 1.3 to 1.0) with ripple opacity.	400 ms	cubic-bezier(0, 0, 0.2, 1)	Confirms satellite packet reception without shifting page layouts.
Card Expansion (Disclosure)	Height and opacity expansion from 0 to target size.	200 ms	cubic-bezier(0.4, 0, 0.2, 1)	Prevents abrupt UI jumps during details discovery.
Status Alert Trigger	Subtle border-color transition from slate to amber or red.	250 ms	ease-in-out	Directs controller focus to emerging precedence conflicts.

Page / Tab Switching	4px vertical slide with opacity fade (0% to 100%).	150 ms	ease-out	Delivers a smooth, native app-like experience.
Data Skeleton Loader	Shimmer gradient moving left-to-right across blank cards.	1.2 s (loop)	linear	Reduces perceived latency during model inference fetches.

6. Detailed Screen Blueprints & Structural Wireframes

6.1 Screen 1: Section Train Controller Operations Console (Desktop)

Designed for divisional train dispatchers who require real-time situational awareness across their assigned section.

- **Top App Bar (Height: 56px, Fixed):**
 - Left: Indian Railways official seal, application title (DTAPE Control Office - Northern Railway / Delhi Division), and active line indicator (Active Corridor: NDLS - CNB).
 - Center: Real-time clock (14:23:18 IST) and satellite network status badge (RTIS Online: 142 Locomotives Active).
 - Right: Search field (Quick Jump: Train # or Station Code), quick filter toggles (All Trains, Delayed Only, Overtakes Pending), and user profile avatar.
- **Left Sidebar (Width: 320px, Collapsible):**
 - Train filter tabs: High Priority (Rajdhani/Vande Bharat), Mail/Express, Freight.
 - Search results showing train cards:
 - Card Top: Train Number (12952), Train Name (MUMBAI TEJAS RAJDHANI).
 - Card Bottom: Current Speed (128 km/h), Delay Badge (+4 min [On Time]), Next Reporting Station (GZB - Ghaziabad).
- **Center Main Work Area (Fluid):**
 - **Top Sub-Header:** Corridor schematic view showing active block sections and current signal states (Green, Double Yellow, Yellow, Red).
 - **Interactive Time-Distance Chart:**
 - Vertical axis: Stations along the corridor arranged by distance (KM).
 - Horizontal axis: Time progression (Past 2 hours to Next 4 hours).
 - Solid Lines: Realized train runs based on RTIS telemetry.
 - Dotted Lines: Predicted dynamic ETA paths generated by the inference engine.
 - Shaded Bands: 80% Conformalized Quantile intervals around predicted lines.
 - Conflict Indicators: Amber alert diamonds where predicted trajectories intersect on single tracks or platform approaches.

- **Right Inspector Drawer (Width: 380px, Slide-out):**
 - Active selection details: Train 12952 MUMBAI RAJDHANI.
 - Kinematics card: Locomotive ID (WAP-7 #30452), Speed (128 km/h), MPS (130 km/h).
 - Prediction explanation: Engineering Allowance used (2.5 min), TSR speed restrictions (None active), Active Headway (4.2 km to preceding 12004).
 - Downstream arrival table: Station codes, scheduled times, dynamic ETAs, and confidence margins ($\pm \text{min}$).

6.2 Screen 2: Station Master & Platform Berthing View (Tablet / Desktop)

Tailored for Station Masters managing platform tracks, rake turnarounds, and yard movements.

- **Top Status Ribbon:**
 - Station code and name (CNB - KANPUR CENTRAL), Total platforms (10), Active tracks occupied (7 / 10).
- **Platform Berthing Grid (10 Rows for Platforms 1 to 10):**
 - Each platform row displays:
 - Current Occupancy: Train currently berthed, scheduled departure time, boarding progress indicator.
 - Next Incoming Train: Train number, originating station, dynamic ETA countdown (*"Arriving in 14 min"*).
 - Platform Conflict Flag: Red badge alert if a delayed train's dynamic arrival window overlaps with an existing platform occupancy.
- **Approach Countdown Drawer (Bottom / Side Pane):**
 - Chronological list of all coaching trains entering the station approach zone (within 30 km).
 - Shows train number, expected home signal arrival time, predicted platform assignment, and throat clearance status.

6.3 Screen 3: Passenger Dynamic Journey Tracking (Mobile Responsive)

Optimized for rail passengers checking train arrivals on smartphones.

- **Header (Height: 52px, Sticky):**
 - Train identification: 12951 NDLS - MMCT Rajdhani Express.
 - Clean back button, journey date pill (28-Sep-2026), and share button.
- **Hero Status Summary Card:**
 - Surface: Pure white, bordered with #E2E8F0.
 - Headline: Dynamic arrival summary (*"Expected at Kota Jn around 18:32"*).
 - Delay Badge: Amber badge showing Running 14 min late.
 - Confidence Window: Subtext displaying Arrival window: 18:29 – 18:36 (80% certainty).
 - Last Updated Badge: Shows time since last satellite sync (*"Updated 12 seconds ago via RTIS satellite"*).
- **Vertical Station Journey Tracker:**
 - Clean vertical line connecting station milestone nodes:
 - **Passed Stations:** Muted slate text, checked circle icon, actual departure timestamp

- logged.
- **Current Train Position:** Illustrated train icon with a subtle pulsing radar ring, showing current speed (112 km/h) and distance to next stop (24 km to Kota Jn).
- **Upcoming Stations:** High-contrast text, clear scheduled arrival time (e.g., 18:18) with strikethrough styling when delayed, and the new dynamic ETA (e.g., 18:32) displayed in bold type.
- **Contextual Delay Explanation (Expandable Sheet):**
 - Explanation card: *"Why is the train delayed?"*
 - Details: *"Delays occurred due to temporary speed restrictions for track renewal work between Gangapur City and Sawai Madhopur. The train is projected to recover 4 minutes before reaching Vadodara."*

7. Component Specifications & Design Tokens

7.1 Reusable UI Component Library

Design System Components	
— Badges & Chips	DelayStatusChip, LiveGPSBadge, ServiceClassPI
— Navigation	TopNavBar, CorridorSelector, StationTimelineStep
— Data Displays	KinematicsCard, ConformalIntervalBar, PlatformGridRow
— Feedback & Controls	ConflictAlertBanner, Re-routeModal, QuickSearchInput

Component 1: DelayStatusBadge

- **Variants:** on-time ($\leq 5 \text{ min}$), minor ($6-15 \text{ min}$), moderate ($16-60 \text{ min}$), critical ($> 60 \text{ min}$).
- **Properties:**
 - Padding: 4px 8px
 - Border Radius: 4px
 - Font: Inter 12px, Weight: 600, Monospace numbers.
 - Border: 1px solid matching semantic border token.
- **Example Output:** Green background (#ECFDF5), green border (#A7F3D0), green text (#065F46), text: On Time (0 min).

Component 2: ConformalIntervalBar

- **Purpose:** Visualizes prediction uncertainty without technical jargon.
- **Structure:**
 - Horizontal bar (8 px height, rounded 4 px).
 - Background track: Neutral gray (#E2E8F0).
 - Shaded interval: Semi-transparent blue highlight (#BFDDBFE) indicating the 10th to 90th percentile arrival window.
 - Median pin: Solid navy vertical line (#0F3875) marking the most probable arrival

timestamp.

- Accessible label below: "Most likely arrival: 17:42 (Range: 17:38 to 17:46)".

Component 3: StationTimelineNode

- **Purpose:** Core element of the vertical passenger tracker and controller station rosters.
- **States:**
 - Passed: Complete dot (#94A3B8), line solid slate.
 - Active / In-Transit: Blue pulsing circle (#1E5AA8), line progress fill.
 - Upcoming: Open circle outline (#CBD5E1), scheduled time cleanly formatted.

8. Accessibility (a11y), Ergonomics, & Error Handling

8.1 Accessibility Standards

- **WCAG 2.1 Level AAA Compliance:** Every text color pairing matches or exceeds required 7:1 contrast levels.
- **Color Blindness Adaptations:** Color is never used as the sole indicator of delay status. Every status badge pairs color with explicit text labels (e.g., [On Time], [Minor Delay], [Severe Delay]) and distinct icon shapes.
- **Keyboard Navigation:** Full support for standard tab navigation, logical focus indicators (2px solid #3B82F6), and clear keyboard shortcuts for operations desks.
- **Screen Reader Optimization:** Dynamic countdowns include descriptive ARIA live region tags:

```
HTML
```

```
<div role="status" aria-live="polite">
```

```
  Train 12952 estimated to arrive at platform 3 at 14 hours 32 minutes, 6 minutes behind
```

```
  schedule
```

```
</div>
```

8.2 Edge Cases and Error States

Edge Case Scenario	UI System Feedback	Actionable Resolution Path
Telemetry Drop / Satellite Loss	Station card shows amber satellite icon with warning label: GPS signal interrupted (Tunnel / Deep Cutting).	Displays estimated position using dead-reckoning extrapolation and lists elapsed time since last sync ("Last confirmed 4 min ago").
Inference Engine Fallback	Status chip displays neutral tag: Kinematic Schedule Estimate (ML Offline).	Seamlessly falls back to Working Time Table physics calculations with an informational tooltip for

		controllers.
Midnight Rollover Crossing	Date transition displayed inline: Arrival: 00:24 IST (+1 Day, 29-Sep).	Prevents passenger confusion on overnight runs with explicit next-day date labels.
Unscheduled Loop Line Halt	Blue notification banner: Train diverted to Loop Line at Subzi Mandi. Precedence overtake in progress.	Controller console shows expected clearance time based on the preceding express train's path.

9. Implementation Reference Architecture (Frontend Tech Stack)

To ensure high performance and smooth UI rendering, the frontend architecture relies on modern, lightweight frameworks:

- **Core Framework:** Next.js (React 19) / TypeScript (Strict Mode).
- **Styling & Design Tokens:** Tailwind CSS configured with custom railway tokens (theme.colors.irNavy, etc.).
- **State Management:** TanStack Query (React Query) for API polling and server state synchronization, combined with lightweight local state stores.
- **Real-Time Data Transport:** Socket.io client / Native WebSockets for live telemetry and countdown tickers.
- **Vector Map & Control Chart Rendering:** Canvas-based rendering using Deck.gl or HTML5 Canvas for smooth 60 FPS time-distance trajectory charts across thousands of data points.
- **Icons:** Lucide-React outline icons.